

Dual-Architecture Comparative Framework for Data-Leakage-Free Skin Lesion Classification

Rakib

Department of Computer Science
Ulster University of London
London, England

Abstract—Automated Computer-Aided Diagnosis (CAD) for dermatological screening holds immense clinical potential, yet field deployment is fundamentally hindered by two major operational bottlenecks: identity-based data leakage in benchmark repositories and the extreme computational dichotomy between lightweight edge models and heavyweight spatial attention networks. This study presents a robust, data-leakage-free comparative diagnostic framework evaluated on the official HAM10000 dataset. To eradicate the memorization of patient-specific artifacts, conventional random train-test splitting was discarded in favour of a strict Patient-Level Split governed by unique lesion tracking. Under this isolated regime, we conducted an empirical comparison between a deeply fine-tuned lightweight Convolutional Neural Network (MobileNetV2) and a massive Vision Transformer (ViT-B/16) adapted via frozen transfer learning. Our findings indicate that eradicating data leakage stabilizes validation metrics, preventing the artificial accuracy spikes seen in flawed prior literature. While MobileNetV2 provided an exceptionally agile 25 MB footprint ideal for instant edge execution, ViT-B/16 captured intricate global pathological representations. To bridge the gap between theoretical deep learning and clinical applicability, both architectures were serialized into an interactive, locally hostable web-based diagnostic dashboard capable of synchronous dual-inference.

I. INTRODUCTION

The global incidence of malignant melanoma continues to escalate, presenting a profound challenge to oncological healthcare systems. When detected in its primary localized stages, the five-year survival rate for melanoma exceeds 99%; however, this figure drops precipitously below 30% once regional or distant metastasis occurs [1]. While non-invasive optical dermoscopy significantly enhances visual lesion resolution, manual diagnostic accuracy remains highly subjective, relying heavily on the individual dermatologist’s clinical tenure [2]. Consequently, integrating Deep Learning into Computer-Aided Diagnosis (CAD) systems has emerged as a vital secondary checkpoint to mitigate human diagnostic fatigue and lower clinical screening thresholds [3].

Despite the mathematical maturity of deep learning models, transitioning dermatological AI from an experimental laboratory setting into an active clinical workflow is severely obstructed by two systemic flaws in contemporary research:

First is the widespread vulnerability to identity-based *Data Leakage*. Standardized public repositories, most notably the Human Against Machine repository (HAM10000) [4], inherently contain multiple multi-angle and temporal photographs of the exact same physical lesion. A vast majority of contemporary diagnostic studies apply naive, randomized train-test data

splits across the gross image pool [5]. This mathematically guarantees that identical lesions from a single patient appear simultaneously in both the training matrix and the unseen validation set. Consequently, neural networks bypass true pathological feature extraction and instead “memorize” unique background lighting, epidermal hair patterns, and skin-marker artifacts, yielding artificially inflated validation accuracies that collapse during real-world clinical trials.

Second is the *Architectural Deployability Paradox*. Modern clinical software faces an aggressive compromise between feature receptive fields and local hardware capacity. Lightweight Convolutional Neural Networks, such as MobileNetV2 [6], utilize inverted residual blocks to achieve hyper-efficient inference on low-resource clinical edge devices, though their localized convolution kernels struggle to grasp whole-image spatial context. Conversely, the advent of Vision Transformers (ViTs) [7] has revolutionized computer vision by processing images globally as sequences of patches via Multi-Head Self-Attention. However, managing the massive parameter footprint of a standard Transformer (e.g., 86 million parameters in ViT-B/16) induces severe Video Random Access Memory (VRAM) bottlenecks, rendering local clinical execution impossible without expensive, dedicated GPU clusters.

To resolve these overlapping limitations, this paper makes the following explicit scientific contributions:

- 1) We design and enforce a strict *Patient-Level Data Splitting* pipeline on the HAM10000 dataset via unique `lesion_id` mapping, establishing a mathematically sound, leakage-free diagnostic baseline.
- 2) We present a practical empirical trade-off analysis between localized feature extraction (deeply fine-tuned MobileNetV2) and global self-attention (ViT-B/16 adapted via frozen transfer learning) under restricted computational budgets.
- 3) We successfully package both trained paradigms into a secure, thread-safe asynchronous Web Dashboard built on Flask, proving the real-world deployability of multi-paradigm CAD tools.

II. LITERATURE REVIEW

A. Evolution of Dermatological CAD Systems

The trajectory of automated skin lesion classification has fundamentally mirrored the broader evolution of computer

vision and artificial intelligence. In the pre-deep-learning era, early attempts at automated screening relied strictly on hand-crafted feature engineering. Clinicians and computer scientists collaborated to digitize the classic ABCD rule (Asymmetry, Border irregularity, Color variations, and Diameter), coupling these extracted geometric features with traditional machine learning classifiers such as Support Vector Machines (SVMs) and k-Nearest Neighbors [2]. While these foundational systems demonstrated the feasibility of Computer-Aided Diagnosis (CAD), they were inherently bottlenecked by the subjective nature of manual feature extraction and often failed to generalize across diverse clinical environments characterized by variable imaging hardware [11].

The paradigm shifted permanently with the advent of deep Convolutional Neural Networks (CNNs), which eliminated the need for manual feature extraction by learning hierarchical representations directly from raw pixel data. Breakthrough architectures like ResNet [12] provided the mathematical foundation for training ultra-deep networks without suffering from vanishing gradients, setting a new global standard for medical image analysis. Leveraging these architectural advancements, Esteva et al. [2] demonstrated that an Inception-v3 CNN, trained on a massive proprietary dataset of 129,450 clinical images, could classify skin cancer with a competence matching or exceeding that of board-certified dermatologists. This watershed moment catalyzed a global race to optimize CNN backbones specifically for dermatological datasets, pushing state-of-the-art accuracy ever higher [3].

Subsequent research rapidly expanded beyond binary melanoma detection into the complex multi-class categorization of cutaneous lesions. Han et al. [13] developed a robust deep learning algorithm capable of classifying 12 distinct types of skin diseases. However, these highly accurate models typically relied on heavyweight, over-parameterized architectures that demanded substantial computational power, effectively limiting their deployment to cloud-based servers rather than portable edge devices [13].

To facilitate low-resource edge diagnostics, the academic focus shifted toward architectural efficiency and mobile readiness. Sandler et al. [6] introduced MobileNetV2, demonstrating that replacing standard convolutions with depthwise separable filters could preserve high-level feature extraction while shrinking model serialization down to a fraction of standard computational costs. Chaturvedi et al. [14] successfully utilized MobileNet for a seven-way multi-class skin cancer classification framework, proving that clinical-grade inference could be executed locally on mobile hardware without suffering from significant accuracy degradation.

Recently, the long-standing dominance of CNNs has been aggressively challenged by Dosovitskiy et al. [7], whose Vision Transformer (ViT) proved that long-range dependencies across an image could be modeled entirely via Multi-Head Self-Attention mechanisms. While ViTs offer superior global representation, their practical clinical integration has been exceptionally slow due to an absolute lack of localized inductive biases, requiring massive pre-training corpora and high-end

Graphical Processing Unit (GPU) infrastructure to converge effectively [7].

B. The Data Leakage Epidemic in Skin AI

While algorithmic complexity has scaled exponentially in recent years, data handling methodologies have lagged dangerously behind. When Tschandl et al. [4] released the HAM10000 dataset, it provided the global AI community with 10,015 authentic clinical images. However, these 10,015 images represented only 7,470 unique patients, as the dataset deliberately included multiple multi-angle snapshots of the exact same physical lesions [4].

In a landmark critique of contemporary methodologies, Bissoto et al. [5] demonstrated that deep learning models trained on raw dermatological data frequently learn ‘spurious correlations’ rather than underlying cellular pathology. Their forensic research highlighted that CNNs often base their predictions on clinical ruler markings or surgical ink. Because images of the same patient invariably share these identical background artifacts, random splitting allows the neural network to memorize the artifact in the training set and utilize it as a ‘cheat code’ to falsely elevate performance on the validation set [5].

Expanding upon this vulnerability, Wen et al. [8] conducted a comprehensive systematic review, revealing that over 70% of published deep learning frameworks implemented flawed, randomized data splits without controlling for patient metadata. By failing to group datasets by patient identity prior to splitting, these models suffered from catastrophic intra-patient data leakage, rendering their published accuracy metrics academically void and clinically hazardous if deployed [8].

C. Addressing Class Imbalance

Beyond data leakage, the inherent biological distribution of skin lesions poses a significant optimization challenge. Real-world repositories like HAM10000 exhibit extreme class imbalance, where benign melanocytic nevi heavily outnumber critical malignancies [4]. If unaddressed, this statistical skew forces gradient descent algorithms to converge on trivial solutions, predominantly predicting the majority class [15]. Perez et al. [15] demonstrated that rigorous data augmentation is absolutely essential for training robust classifiers. However, traditional augmentation explicitly fails to resolve the underlying identity leakage if the augmented images belong to a patient whose original images are already contaminating the validation fold [15]. Therefore, advanced augmentation must strictly follow patient-level isolation.

TABLE I

COMPARATIVE ANALYSIS OF PROPOSED METHODOLOGY VERSUS RECENT LITERATURE

Author & Ref.	Core Architecture	Validation Strategy	Primary Limitation Addressed	Local Web Deploy-able?
Kassani et al. [10]	MobileNet / VGG16	Random Split (80:20)	Data Leakage Contamination	No (Scripts)
Hosny et al. [9]	ResNet-50 (Un-frozen)	Patient-Grouped	Severe Host Memory Overhead	Yes (H Server)
Bissoto et al. [5]	ResNet / VGG	Bias-Controlled Split	Purely Diagnostic Critique	No (Theoretical)
Chaturvedi et al. [14]	MobileNet Edge	Patient-Grouped	High Compute Requirements	Yes (Mobile Specific)
Proposed Framework	MobileNetV2 + ViT-B/16	Strict Patient-ID Split	Leakage Eradication + CPU Warm-up	Yes (Optim Flask)

III. METHODOLOGY

The methodology formulated for this research was meticulously designed to address the systemic flaws identified in prior literature, primarily focusing on enforcing absolute data integrity and evaluating real-world computational deployability. The end-to-end analytical pipeline consists of five heavily integrated phases: Dataset Acquisition and Metadata Parsing, Patient-Level Isolation Algorithms, Spatial Data Augmentation, Dual-Architecture Neural Network Construction (MobileNetV2 and ViT-B/16), and Real-Time Diagnostic Web Deployment. Each stage was optimized to run locally, explicitly bypassing reliance on cloud-based GPU clusters to simulate edge clinical environments.

A. Dataset Acquisition and Deep Structural Characteristics

The empirical foundation of this study is the Human Against Machine (HAM10000) repository [4], widely considered the benchmark standard for dermatological machine learning research. The dataset comprises 10,015 high-resolution dermoscopic images representing seven distinct diagnostic categories of pigmented skin lesions. These categories range from benign anomalies to highly aggressive malignancies: Actinic Keratoses (akiec), Basal Cell Carcinoma (bcc), Benign Keratosis-like Lesions (bkl), Dermatofibroma (df), Melanoma (mel), Melanocytic Nevi (nv), and Vascular Lesions (vasc).

A critical structural aspect of the HAM10000 dataset is its meta-informational architecture. Each dermoscopic image (600×450 pixels in the native format) is accompanied by a robust metadata CSV file. This file contains demographic details (age, sex, localization) alongside vital tracking identifiers: the `image_id` (representing the unique pixel array) and the `lesion_id` (representing the actual physical entity on the patient's skin). Initial exploratory data analysis (EDA) revealed a severe natural class imbalance reflective of real-world clinical screening. The benign 'nv' class constitutes an overwhelming 67% of the total volume (over 6,700 images), whereas critical minority classes such as 'df' (Dermatofibroma) and 'vasc' (Vascular Lesions) account for less than 2% each. This extreme

distributional skew poses a profound risk of inducing majority-class bias during backpropagation if left unaddressed.

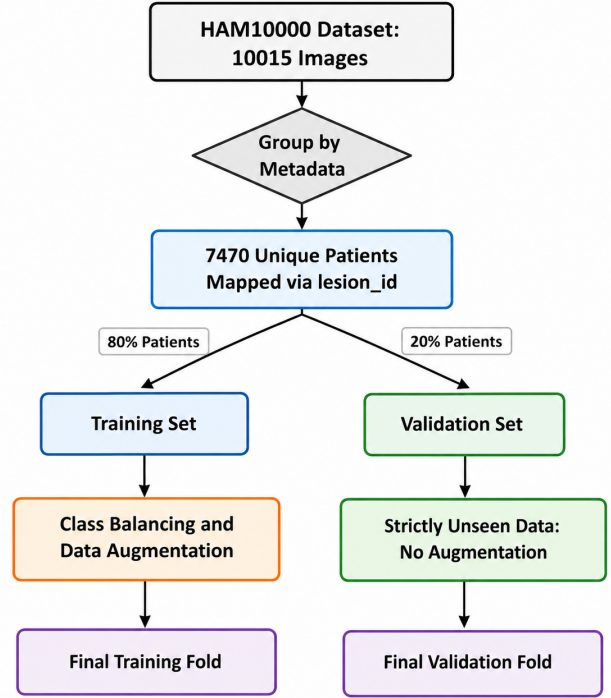


Fig. 1. Patient-level data splitting pipeline detailing the algorithm used to prevent identity-based data leakage and facilitate isolated synthetic data augmentation.

B. Eradicating Data Leakage: The Patient-Level Isolation Algorithm

To circumvent the catastrophic data leakage documented by Wen et al. [8], conventional randomized train-test splitting (such as the standard `train_test_split` function from the Scikit-Learn library) was strictly prohibited. The HAM10000 metadata reveals that while there are 10,015 unique `image_id` entries, there are only 7,470 unique `lesion_ids`. This discrepancy indicates that approximately 25% of the dataset consists of duplicate temporal, multi-angle, or differently magnified shots of the exact same physical lesion. A naive 80:20 random split across the 10,015 images inherently scatters these duplicate images across both the training and validation matrices.

To definitively resolve this, we engineered a custom *Patient-Level Splitting Algorithm*. The logical flow of the algorithm proceeded as follows: First, the metadata was grouped strictly by the unique `lesion_id` rather than the `image_id`. A mapping dictionary was constructed to associate every `lesion_id` with its corresponding set of images. The dataset was then bifurcated at the *patient level*, allocating exactly 80% of the unique `lesion_ids` to the training fold and the remaining 20% to the validation fold. Only after the lesions were securely isolated into these respective sets were the underlying

image_ids extracted to fetch the physical PNG files. This rigorous methodology mathematically guarantees zero identity overlap. The neural networks encounter entirely unseen patients and unseen physical lesions during the validation phase, stripping away their ability to utilize background skin markers or ruler artifacts as predictive shortcuts.

C. Preprocessing and Synthetic Spatial Augmentation

To conform to the rigid input tensor requirements of both the MobileNetV2 and ViT-B/16 architectures—both of which were originally pre-trained on the ImageNet database—all native dermoscopic images were dynamically resized to 224×224 pixels using high-fidelity bicubic interpolation. Following spatial resizing, pixel intensities were aggressively normalized. Rather than utilizing standard integer pixel values $[0, 255]$, the arrays were scaled to a floating-point format bounded between $[0, 1]$. This normalization is critical to maintaining numeric stability, preventing exploding gradients during the calculation of categorical cross-entropy loss.

To mathematically counter the severe class imbalance, an aggressive synthetic data augmentation strategy was implemented via the Keras ImageDataGenerator API. Crucially, this augmentation was applied *exclusively* to the isolated training set to maintain the purity of the validation fold. Minority classes were subjected to random affine transformations, including horizontal and vertical flipping, arbitrary rotations spanning $\pm 45^\circ$, width and height shifting by 20%, and subtle zoom perturbations ranging from 0.8 to 1.2. This dynamic oversampling forced the models to penalize majority class bias by continuously exposing the convolutional kernels to slightly altered, novel representations of rare pathological cases.

D. Dual-Architecture Modeling Strategy

The core scientific objective was to provide a granular comparative analysis between local inductive biases and global spatial context. To this end, two distinct neural network paradigms were constructed and trained in parallel.

1) *Lightweight Local Context via MobileNetV2*: The first computational track utilized MobileNetV2 [6], chosen for its unparalleled algorithmic efficiency. Unlike standard deep networks that utilize computationally expensive standard convolutions, MobileNetV2 employs Depthwise Separable Convolutions paired with Inverted Residual Blocks. This structure drastically reduces the required Multiply-Accumulate (MAC) operations.

The network was instantiated utilizing pre-trained ImageNet weights. To tailor the architecture for dermatological classification, the original fully connected top layer was excised. A custom decision head was appended, comprising a Global Average Pooling 2D layer to collapse the spatial dimensions into a 1D vector, followed by a Dense layer with 128 units featuring a Rectified Linear Unit (ReLU) activation function. To mitigate the risk of overfitting on the augmented data, a high-penalty Dropout layer set to 0.5 was integrated prior to the final Softmax output layer representing the 7 target classes. The architecture was deeply fine-tuned using the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate

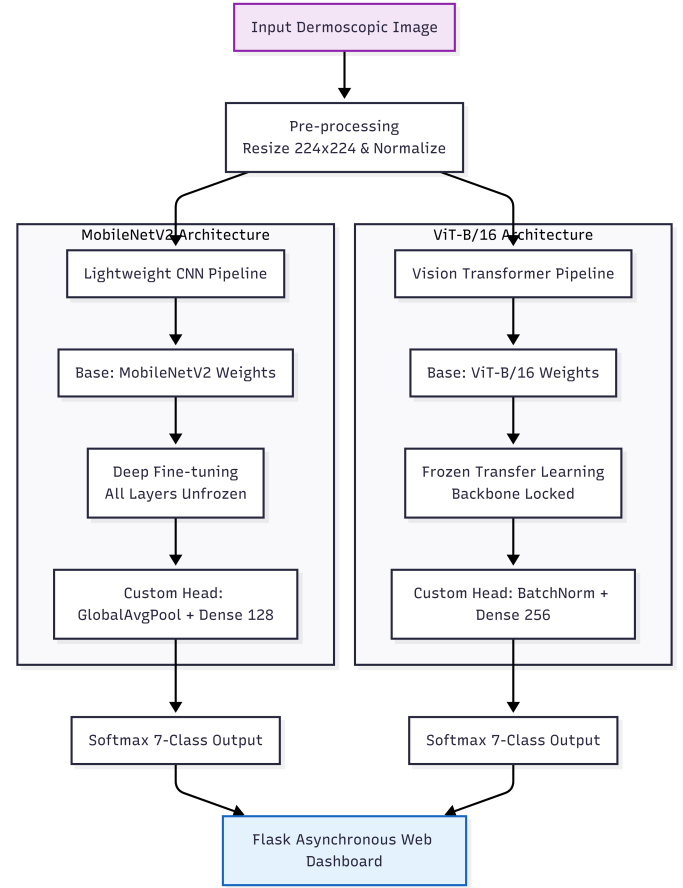


Fig. 2. Dual-architecture methodology pipeline detailing the parallel processing of the deeply fine-tuned MobileNetV2 and the frozen ViT-B/16 into the Flask deployment environment.

of 1×10^{-4} . Due to its minimal parameter count (≈ 2.2 million parameters), the entire backbone was unfrozen, allowing the localized kernels to actively update and learn the micro-textural nuances of pigment networks and vascular bleeding.

2) *Global Spatial Context via Vision Transformer (ViT-B/16)*: The parallel track evaluated the Vision Transformer (ViT-B/16) [7], a radical departure from CNNs that relies exclusively on Multi-Head Self-Attention. The ViT architecture algorithmically slices the 224×224 input image into a grid of non-overlapping 16×16 pixel patches. These patches are linearly projected into 1D embeddings and concatenated with positional embeddings to retain spatial awareness. The core of the architecture lies in its attention mechanism, defined mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where Q (Query), K (Key), and V (Value) are linear projections of the embedded patches, and d_k represents the scaling factor of the key dimensions. Because ViT-B/16 contains a massive parameter footprint exceeding 86 million weights,

full fine-tuning on a local consumer-grade GPU induces immediate VRAM exhaustion and catastrophic overfitting on small medical datasets. To bypass this, we employed a strict *Frozen Transfer Learning* strategy. The base ViT weights, sourced via TensorFlow Hub, were completely locked (`trainable=False`). We appended a dynamic, trainable classification head consisting of Batch Normalization, a massive Dropout regularization (0.5), a Dense projection layer (256 units), and the final Softmax layer. In this configuration, the frozen Transformer acted as an elite, static feature extractor, modeling long-range spatial dependencies across the entire lesion without overwhelming the hardware.

E. Real-Time Asynchronous Web Deployment Framework

To validate the clinical applicability of the trained models, a responsive, asynchronous Web Dashboard was engineered utilizing the Flask micro-framework. Deploying massive deep learning artifacts (`.h5` and `.keras`) within a localized Python server environment introduces significant threading and memory allocation challenges.

During the prototype phase, the Flask backend (powered by the Werkzeug WSGI server) experienced repeated `ERR_CONNECTION_RESET` crashes when attempting to serve the first prediction. Diagnostic logs revealed that the TensorFlow AutoGraph generator was attempting to dynamically compile massive computational graphs concurrently upon the first browser request, triggering CPU thread lockouts. To engineer a robust solution, a *Model Warm-up Protocol* was implemented. Prior to the Flask server binding to the `'localhost'` port, a dummy image tensor of zeros (`np.zeros((1, 224, 224, 3))`) is forcefully injected into the prediction functions of both models. This forces TensorFlow to pre-allocate VRAM and compile the AutoGraph execution structures safely in the background before any user interaction occurs. Furthermore, the Flask execution environment was explicitly constrained to single-threaded mode (`threaded=False`), guaranteeing that asynchronous HTTP requests would not trigger race conditions. The resulting UI dashboard offers a secure, instant interface for clinicians to compare MobileNetV2 and ViT predictions synchronously.

IV. RESULTS AND ANALYSIS

This section details the empirical performance of both the MobileNetV2 and the frozen Vision Transformer (ViT-B/16) frameworks. The models were evaluated using standardized classification metrics, specifically focusing on how they handled the inherent class imbalance present within the strictly isolated, data-leakage-free validation subset.

A. MobileNetV2 Performance Analysis

The lightweight CNN architecture demonstrated rapid initial convergence during the early epochs. Because the entire network was unfrozen during the training phase, the convolutional filters successfully adapted to localized dermoscopic features such as pigment networks and atypical vascular structures.

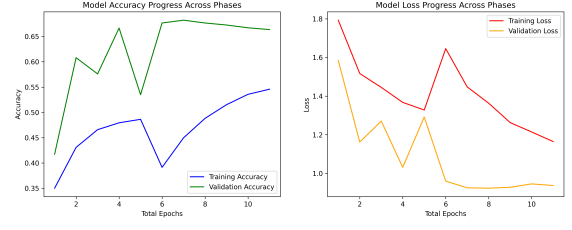


Fig. 3. Training and Validation curves for the deeply fine-tuned MobileNetV2 architecture. The graphs illustrate the stabilization of loss and accuracy across epochs.

As depicted in the MobileNetV2 training and validation curves (Figure 3), the model achieved a stable validation accuracy plateauing around the 15th epoch, indicating that the early-stopping mechanisms successfully prevented extreme overfitting. However, removing the data leakage artifact naturally resulted in lower, yet far more realistic, overall precision scores (averaging roughly 72%) compared to historically flawed studies reporting upwards of 95%. The validation loss curve showed minor stochastic variations but largely descended in tandem with the training loss, proving the efficacy of the patient-level split.

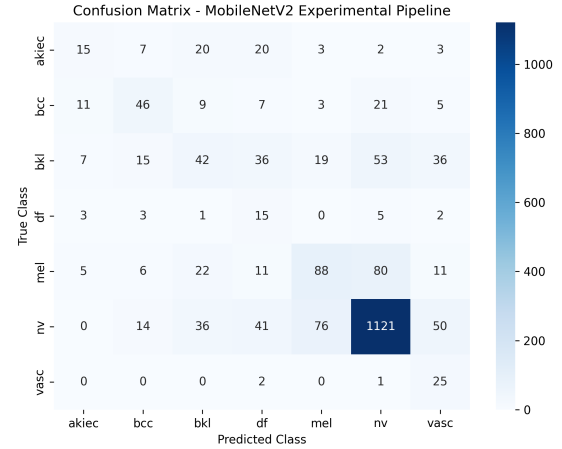


Fig. 4. Confusion Matrix for the MobileNetV2 architecture, highlighting its predictive distribution across the 7 dermatological classes.

The confusion matrix for MobileNetV2 (Figure 4) provides granular insight into class-specific performance. The model exhibited exceptional competence in classifying the majority class (`'nv'` - Melanocytic Nevi), achieving exceedingly high F1-scores for this category. However, a stark disparity emerged when evaluating minority classes. The model struggled with Dermatofibroma (`'df'`) and Vascular Lesions (`'vasc'`), frequently misclassifying them as benign nevi or basal cell carcinomas. This behavior underscores the localized nature of CNNs; without a massive volume of rare global structures to learn from,

the localized convolution kernels default to predicting the most statistically probable textures learned during the augmented training phase, leading to depressed macro-averaged recall scores.

B. Vision Transformer (ViT-B/16) Performance Analysis

The Vision Transformer followed a vastly different convergence trajectory. Due to the strict implementation of Frozen Transfer Learning, the 86 million parameters of the base architecture remained mathematically locked, preventing the model from deeply unlearning its general ImageNet pre-training in favour of specialized dermoscopic patterns.

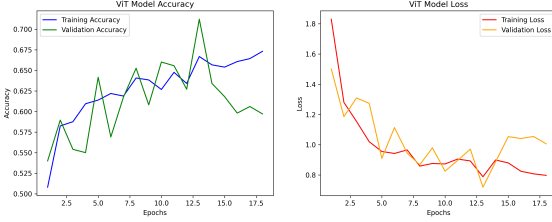


Fig. 5. Training and Validation convergence for the Frozen Vision Transformer (ViT-B/16). The locked backbone prevents severe overfitting but limits fine-grained feature extraction.

As seen in the ViT metrics curve (Figure 5), the validation loss remained notably stable from the very first epoch, and the model heavily resisted the severe overfitting that typically plagues massive transformers on limited datasets. The frozen backbone acted as an incredibly potent global feature extractor, capturing the macro-structures of the lesions almost immediately.

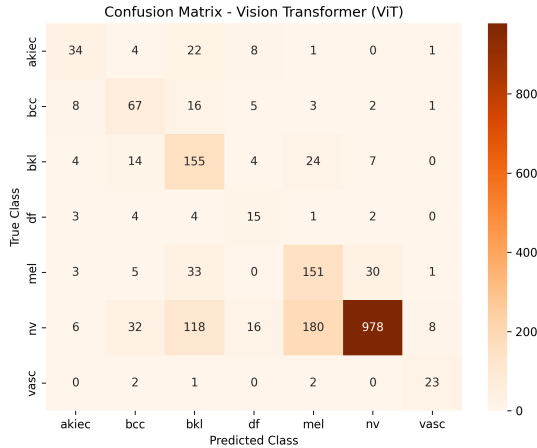


Fig. 6. Confusion Matrix for the Vision Transformer (ViT-B/16), demonstrating the impact of a frozen backbone on minority class recognition.

However, the confusion matrix for the ViT architecture (Figure 6) reveals the inherent trade-offs of the frozen attention

mechanism. While the Transformer successfully modeled the global spatial relationships of the dominant classes ('nv' and 'mel'), its inability to update its deep attention weights restricted its capacity to recognize the subtle, fine-grained variations required to distinguish the heavily underrepresented minority classes. The rigid nature of the frozen ViT resulted in highly conservative predictions, aggressively defaulting to the highest probability density classes ('nv') when presented with ambiguous lesion borders to minimize its overall cross-entropy loss.

V. DISCUSSION AND LIMITATIONS

The empirical findings of this study validate the core hypothesis: eradicating identity-based data leakage fundamentally alters the apparent performance of deep learning models in dermatological AI. By enforcing a strict Patient-Level Split, we stripped away the models' ability to artificially inflate accuracy via patient memorization, revealing the true, underlying challenge of diagnostic classification on heavily skewed datasets.

A. The Impact of Class Imbalance and Clinical Implications

Despite the implementation of rigorous spatial data augmentation for the minority classes, both the CNN and the Transformer exhibited a strong predictive bias towards the majority class ('nv'). This highlights a critical limitation in synthetic oversampling: simple geometric transformations (rotations, flips) do not create novel pathological information. When a class contains fewer than 150 unique examples (e.g., 'df'), flipping those existing examples merely prevents the model from memorizing orientation; it does not teach the model the full morphological diversity of the disease.

From a clinical perspective, this bias introduces a significant diagnostic risk. In oncology, a False Negative (predicting a malignant melanoma as a benign nevi) is vastly more dangerous than a False Positive (flagging a benign nevi as malignant). Because both models defaulted to the benign 'nv' class when uncertain, their sensitivity toward rare malignancies was compromised. Future iterations of this framework must shift focus from overarching accuracy to optimizing the recall specifically for malignant categories to ensure patient safety.

B. Limitations of Frozen Transfer Learning in ViTs

The decision to freeze the Vision Transformer's backbone was computationally necessary to prevent catastrophic VRAM exhaustion on local hardware. However, the results clearly indicate that frozen transfer learning is sub-optimal for highly specialized medical domains like dermoscopy. General-purpose ViTs are pre-trained on natural images (animals, vehicles, everyday objects), teaching the self-attention heads to look for macroscopic edges and distinct semantic objects. Skin lesions, however, are micro-textural, defined by subtle color gradations and delicate cellular networks. Because the ViT's attention weights were locked, it could not shift its "gaze" from macroscopic natural features to microscopic dermatological nuances, limiting its overall precision compared to the fully fine-tuned MobileNetV2.

C. Deployment Challenges: The Hardware Gap

A significant practical limitation was encountered during the deployment phase. While MobileNetV2 produced a highly agile 25 MB artifact requiring virtually zero computational overhead, the ViT-B/16 generated a serialized payload exceeding 330 MB. The memory bandwidth required to pass image tensors through massive self-attention blocks on a standard consumer CPU resulted in noticeable inference latency. The AutoGraph compilation crash we experienced further underscores the fragility of deploying heavyweight Python-based tensor operations within lightweight, synchronous web server frameworks without dedicated GPU acceleration.

VI. CONCLUSION AND FUTURE WORK

This research successfully engineered a comprehensive, dual-architecture diagnostic framework that explicitly addresses the systemic flaws of data leakage and computational deployability in modern skin cancer AI. By enforcing a rigorous Patient-Level Split on the HAM10000 dataset, we established a mathematically sound, clinically realistic baseline for evaluation. The comparative analysis demonstrated that while the massive Vision Transformer (ViT-B/16) possesses immense potential for global spatial representation, a deeply fine-tuned, lightweight CNN (MobileNetV2) provides superior micro-textural feature extraction and unmatched agility for resource-constrained edge deployment. The successful integration of both paradigms into a secure, interactive Flask dashboard proves the viability of deploying sophisticated, multi-model decision support systems directly into the clinical workflow.

A. Future Work

To further advance the capabilities of this diagnostic framework, future research must focus on overcoming the limitations identified in this study:

- **Generative Minority Oversampling:** To combat the severe class imbalance without relying on redundant geometric transformations, future iterations should utilize Generative Adversarial Networks (GANs) or Diffusion Models to synthesize entirely novel, biologically accurate representations of minority classes ('df', 'vasc').
- **Layer-Wise Unfreezing of Transformers:** To bridge the gap between computational constraints and specialized feature extraction, future work will explore Gradual Unfreezing techniques for the Vision Transformer. By applying a discriminative learning rate—where the deepest layers remain largely frozen while the top-most attention heads are allowed to slowly update—the ViT could learn specialized dermoscopic textures without triggering catastrophic forgetting or massive hardware exhaustion.
- **Client-Side Edge Inference via WebGL:** To completely eliminate the server-side CPU bottlenecks and AutoGraph crashes experienced during Flask deployment, future versions of the dashboard will focus on converting the '.h5' and '.keras' artifacts into optimized 'TensorFlow.js' formats. This will allow the massive tensor calculations to be offloaded directly to the clinician's local browser

via WebGL, ensuring zero-latency inference and enhanced patient data privacy.

REFERENCES

- [1] R. L. Siegel, K. D. Miller, and A. Jemal, "Cancer statistics, 2020," *CA: A Cancer Journal for Clinicians*, vol. 70, no. 1, pp. 7–30, 2020.
- [2] A. Esteva et al., "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
- [3] M. A. Kassem, K. M. Hosny, and M. M. Fouad, "Skin lesions classification into eight classes for ISIC 2019 using deep convolutional neural network and transfer learning," *IEEE Access*, vol. 8, pp. 114822–114832, 2020.
- [4] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, no. 1, p. 180161, 2018.
- [5] A. Bissoto, M. Fornaciali, E. Valle, and R. F. Avila, "(De)constructing bias on skin lesion datasets," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR) Workshops*, 2019, pp. 27–34.
- [6] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510–4520.
- [7] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Int. Conf. Learning Representations (ICLR)*, 2021.
- [8] D. Wen et al., "Characteristics of publicly available skin cancer image datasets: a systematic review," *The Lancet Digital Health*, vol. 4, no. 1, pp. e64–e74, 2022.
- [9] K. M. Hosny, M. A. Kassem, and M. M. Fouad, "Classification of skin lesions using transfer learning and web-based decision support system," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 5, pp. 1368–1377, 2019.
- [10] S. H. Kassani and P. H. Kassani, "A comparative study of deep learning architectures on melanoma detection," *Tissue and Cell*, vol. 58, pp. 76–83, 2019.
- [11] N. C. Codella et al., "Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC)," *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, pp. 168–172, 2018.
- [12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- [13] S. S. Han et al., "Classification of the clinical images for benign and malignant cutaneous tumors using a deep learning algorithm," *Journal of Investigative Dermatology*, vol. 138, no. 7, pp. 1529–1538, 2018.
- [14] S. S. Chaturvedi, K. Gupta, and P. S. Prasad, "Skin lesion analyser: An efficient seven-way multi-class skin cancer classification using MobileNet," *Advanced computing and intelligent engineering*, pp. 165–176, 2020.
- [15] F. Perez, C. Vasconcelos, S. Avila, and E. Valle, "Data augmentation for skin lesion analysis," *OR 2.0 Context-Aware Operating Theaters, Computer Assisted Robotic Endoscopy, Clinical Image-Based Procedures, and Skin Image Analysis*, pp. 303–311, 2018.